

The greenhouse gas cost of irrigation as adaptation

Bonnie M. McGill

 Check for updates

Energy use from on-farm pumping only captures two-thirds of US greenhouse gas emissions from irrigation.

Irrigation is an important adaptation measure for growing crops in increasingly hotter and drier growing seasons. Already about 25% of global cropland is irrigated, supporting about 33% of global crop production¹. Ninety per cent of global consumptive water use is for irrigation, half of which is supplied by groundwater². Traditionally, irrigation has enabled crop production in arid areas. Today, irrigation is expanding in humid regions where heat is increasing, rainfall is less reliable, and droughts develop more quickly. In the US, irrigation has expanded at about 70,000 hectares per year in the more humid east in contrast with reductions in irrigated areas in the western US in places like California, Colorado, and Utah due to water scarcity³. Adaptation with irrigation comes at a cost not only in terms of water use but also greenhouse gas emissions. One-third of global anthropogenic greenhouse gas emissions already come from food systems. Given the expanding role of irrigation in buffering the global food supply from the changing climate and the urgent need for the agricultural sector to meet emission reduction targets, it is critical to understand how much irrigation contributes to multiple sources of greenhouse gas emissions at the local level.

Now, writing in *Nature Water*, Avery Driscoll and colleagues provide a first look at county-level irrigation-related greenhouse gas emissions from multiple sources for the contiguous US⁴. The majority of studies assessing greenhouse gas emissions from irrigation have focused on on-farm energy use for pumping water. However, irrigation impacts at least three other greenhouse gas producing processes as well. First, irrigation in arid areas like the western US can rely on inter-basin transfers of water, which often require energy to deliver. The authors built a first-of-its-kind dataset on the energy used by these projects and the proportion of water used by agriculture by contacting project operators. Second, groundwater is often supersaturated with gases that have dissolved into the water along its flowpath, which degas to equilibrate with the atmosphere when brought to the surface. Driscoll and colleagues estimated CO₂ degassing from groundwater used for irrigation using water chemistry and land surface-temperature data. Third, wetting nitrogen-rich soils tends to increase the potential for denitrification, which can produce nitrous oxide (N₂O). The authors used an ecosystem biogeochemical model (DayCent) to estimate the impact of irrigation on N₂O emissions. This work then incorporates county-level estimates of on-farm energy use for irrigation from an article published earlier this year by Driscoll and colleagues in *Nature Communications*⁵.

They estimate that irrigation in the US for the years 2015–2022 led to 18.9 million metric tonnes of CO₂ equivalent (CO₂e) per year with 67% of that from on-farm energy used for pumping irrigation water,



15% from increased N₂O emissions from soils, 13% from CO₂ degassing from groundwater, and 6% from off-farm energy used for inter-basin transfers of water. Meaning, previous estimates of greenhouse gas emissions from irrigation incorporating only on-farm energy use may have only captured two-thirds of emissions. Driscoll and colleagues found similar volumes of groundwater and surface water used for irrigation nationally, but irrigation with groundwater made up 79% of emissions due to greater energy required for on-farm pumping and degassing.

Irrigation-related emissions are highly skewed across US counties with implications for mitigation actions. The authors found hotspots in the High Plains Aquifer region, the Mississippi Delta, California's Central and Imperial valleys, and Southern Arizona. As expected, counties with greater irrigated area tended to have greater county-level irrigation-related emissions; however spatial variability was also driven by the per-hectare (ha) emissions intensity. Counties in the top 10% of the emissions distribution averaged 1.0 tonne CO₂e per ha, whereas the remaining counties averaged 0.5 tonne CO₂e per ha. The highest emitting counties tended to be those with greater water use per ha and found in more arid areas that rely on groundwater and an interbasin transfer.

Mitigation activities reducing emissions from on-farm energy use would target the dominant source of emissions in 90% of counties. The other 10% of counties are dominated by N₂O emissions (where average N application rates are higher and precipitation-to-evapotranspiration ratios are lower, such as Texas, Kansas, and California), degassing from groundwater (where groundwater has higher partial pressure of CO₂ such as the Mississippi River Valley Alluvial Aquifer), or inter-basin transfers (projects that involve energy use and large transfer volumes, elevation gains, and conveyance distances, most notably the Central Arizona Project and California State Water Project).

This work has several important uses. First, the methodology can be used in the US Greenhouse Gas Inventory to improve estimations of greenhouse gas emissions from irrigation as the footprint of irrigation contracts in some places and expands in others. Second, the dataset will improve life cycle assessment models of irrigated crop products as well as livestock products where livestock are fed irrigated crops such as alfalfa. Third, the dataset can help policymakers weigh the greenhouse gas costs of incentivizing irrigation expansion and improvements in efficiency. It is important to note that improving irrigation efficiency benefits farmers but, paradoxically, rarely reduces overall water withdrawals, such that improving irrigation efficiency does not benefit the public good. This has been observed around the world and is due to the increased efficiency reducing recoverable (and reusable) return flows in the watershed and producers may switch to more water-intensive crops and/or expand the area they are irrigating⁶. Policy to regulate agricultural activities is politically challenging. But an innovative groundwater conservation programme in the High Plains Aquifer region of Kansas is irrigator-driven with active monitoring and regulatory oversight and has demonstrated reduced water use, stabilizing groundwater levels, and farmer profitability⁷. In the Kansas case, producers reduced groundwater pumping mainly by improving irrigation efficiency without switching crops or expanding their irrigated area and in some cases switched to crops with lower water demand.

The work by Driscoll and colleagues points to how existing climate change mitigation strategies have knock-on effects for reducing

irrigation-related emissions. Decarbonizing the electrical grid will reduce irrigation pumping emissions (69% of which are from electricity⁵). This also suggests that where irrigation expands policymakers should incentivize irrigators to use electrical pumps. As new areas look to irrigation for adaptation, Driscoll and colleagues' work provides an important tool for planning irrigation development in ways that minimize the multiple sources of irrigation-related greenhouse gas emissions.

Bonnie M. McGill  

American Farmland Trust, Washington, DC, USA.

✉ e-mail: bmcgill@farmland.org

Published online: 11 September 2024

References

1. Portmann, F. T., Siebert, S. & Döll, P. *Glob. Biogeochem. Cycles* **24**, 2008GB003435 (2010).
2. Siebert, S. et al. *Hydrol. Earth Syst. Sci.* **14**, 1863–1880 (2010).
3. Xie, Y. & Lark, T. J. *Remote Sens. Environ.* **260**, 112445 (2021).
4. Driscoll, A. W. et al. *Nat. Water* <https://doi.org/10.1038/s44221-024-00283-w> (2024).
5. Driscoll, A. W., Conant, R. T., Marston, L. T., Choi, E. & Mueller, N. D. *Nat. Commun.* **15**, 675 (2024).
6. Grafton, R. Q. et al. *Science* **361**, 748–750 (2018).
7. Deines, J. M., Kendall, A. D., Butler, J. J. & Hyndman, D. W. *Environ. Res. Lett.* **14**, 044014 (2019).

Competing interests

The author declares no competing interests.